

Advanced Proofs and Mathematical Writing

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Chapter 1

Foundations

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1.1 Notation and conventions

We use standard notation: $\mathbb{N}$ for natural numbers, $\mathbb{R}$ for real numbers, etc.

1.2 Sample theorem and proof types

Theorem 1.1 (Basic inequality). Let $a, b \in \mathbb{R}$. Then $(a - b)^2 \geq 0$ and hence $a^2 + b^2 \geq 2ab$. $\square$

Proof. Expand $(a - b)^2$ to get $a^2 - 2ab + b^2 \geq 0$, rearrange to obtain the result. $\square$

Theorem 1.2. If n^2 is even then n is even. $\square$

Proof by contrapositive. We prove the contrapositive: if n is odd then n^2 is odd. Write $n = 2k + 1$ and compute $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is odd. $\square$

Theorem 1.3. There are infinitely many primes. $\square$

Proof by contradiction. Assume finitely many primes $p_1, \dots, p_k$. Consider $N = \prod_{i=1}^k p_i + 1$. Then N is not divisible by any p_i , contradiction. $\square$

Theorem 1.4. For all $n \in \mathbb{N}$, $\sum_{k=1}^n k = n(n+1)/2$. $\square$

Proof by induction. Base $n = 1$ holds. Assume true for n , then for $n + 1$:

$$\sum_{k=1}^{n+1} k = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}.$$

Thus holds for all n . $\square$

Theorem 1.5. Let $x \in \mathbb{R}$. Then $x^2 \geq 0$. $\square$

Proof. We consider cases.

- If $x \geq 0$ then $x^2 \geq 0$ trivially.
- If $x < 0$ then let $y = -x > 0$. Then $x^2 = y^2 \geq 0$.

$\square$

Theorem 1.6. There exists an irrational number x such that x^x is rational. $\square$

Constructive proof. Consider $x = \sqrt{2}$. If $\sqrt{2}^{\sqrt{2}}$ is rational we are done. Otherwise let $y = \sqrt{2}^{\sqrt{2}}$ which is irrational; then $y^{\sqrt{2}} = 2$ is rational. $\square$

Theorem 1.7. A sample boxed proof $\square$

This is a proof placed inside a shaded box for emphasis. Use this style for long or important proofs.

Chapter 2

Algorithms and Computation

2.1 Pseudocode example

Algorithm 1 Euclid's algorithm for gcd

```
1: procedure GCD( $a, b$ )  
2:   while  $b \neq 0$  do  
3:      $t \leftarrow b$   
4:      $b \leftarrow a \bmod b$   
5:      $a \leftarrow t$   
6:   end while  
7:   return  $a$   
8: end procedure
```

Chapter 3

Graphs and Plots

3.1 TikZ graph example

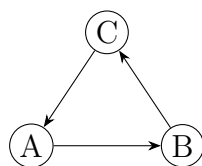


Figure 3.1: Directed triangle

3.2 pgfplots example

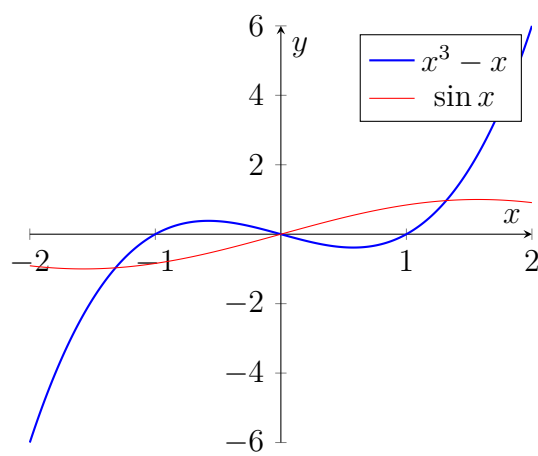


Figure 3.2: Sample plots

Chapter 4

Advanced Math Typesetting

4.1 Commutative diagrams and special symbols

Use `\mathscr` for script letters and `\llbracket\rrbracket` from `stmaryrd` for interval notation.

$$\mathscr{F} : 0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$$

$$\llbracket a, b \rrbracket = \{x \in \mathbb{Z} : a \leq x \leq b\}$$

Appendix A

Appendix

A.1 Useful identities

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}.$$

Exercises Solutions